Kevin Siegall (They/Them)

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EDUCATION

Worcester Polytechnic Institute — **May 2025**
B. S. Robotics Engineering; B. S. Computer Science — Worcester, MA
- Relevant Classes: *Deep Learning For Perception, Swarm Intelligence, AI for Robotics, Unified Robotics: Manipulation, Navigation*

WORK EXPERIENCE

Smartapp.com – R&D Division — **May 2024 – Aug 2024**
Autonomy Engineering Intern — *Worcester, MA*
- Expanded on a Python and React-TS thrust test stand hosted on a Raspberry Pi, with customizable datalogging and test procedures
- Developed an intuitive and flexible motor library, which enables 'hot-swapping' of intelligent motor classes and objects
- Optimized proprietary robot locomotion techniques in Nvidia's IsaacSim using deep reinforcement learning (PyTorch and PPO)

OpenSTEM: Experiential Robotics Platform (XRP) — **Aug 2022 – Present**
Lead Software Developer, XRPLib — *Worcester, MA*
- Designed and maintained a modular MicroPython-based robotics control library deployed across 20K+ classroom-scale robots
- Managed a team of 1-5 over 2.5 years while coordinating with corporate partners and engineers from DEKA and Sparkfun

Worcester Polytechnic Institute — **Oct 2022 - Dec 2022**
Undergraduate Student Assistant — *Worcester, MA*
- Taught algorithm design (Djikstra) and code debugging skills while leading lab sections for a new course (RBE 100x) using XRP robots

Jacobs Technology – Jacobs Software Engineering Center — **May 2022 – Aug 2022**
Software Engineering Intern — *Hudson, NH*
- Refactored and extended C#/.NET mission support tooling to manage dynamic SQL filter sets for DAFIF airspace pathing datasets

PROJECTS

Terrawarden Drone Cleanup – Major Qualifying Project — **Aug 2024 – May 2025**
- Developed aerial manipulator for autonomous waste collection using Intel RealSense and YOLOv11 with low-latency inference (<3ms)
- Created ROS2 Nodes for interfacing with PX4 with code running on a Jetson Orin NX, with Dynamixel actuators controlling the arm
- Implemented object detection model trained on procedurally rendered Blender dataset, enabling robust generalization to roadside litter
- Architected ROS2-based state machine to isolate high-level behavior, programmed using Test-Driven Development (PyTest)

Drone Racing Gate Semantic Segmentation — **Aug 2024 – Oct 2024**
- Used Blender to generate a dataset of 5000 images of drone racing gates for training a semantic segmentation model (U-Net)
- Applied various image augmentation techniques, such as gaussian blur and color jitter, to increase the robustness of the trained model

Video Game AIs — **Jan 2024 – Mar 2025**
- Benchmarked the abilities of an algorithmic model (Expectiminimax) vs a reinforcement model (Q-Learning) at playing Bomberman
- Implemented imitation learning on a Deep Q-Learning model in Pytorch, training it to play the Snake Game

Robotic Navigation – SLAM and AMCL — **Oct 2023 – Dec 2023**
- Developed a robot that could autonomously navigate and map an unknown space, then localize itself when relocated at a later time
- Implemented Simultaneous Localization and Mapping (SLAM), AMCL, A*, and Pure Pursuit on a Turtlebot3 with a planar LiDAR

Hospital Wayfinding & Facilities System — **Mar 2023 – May 2023**
- Developed JavaFX-based kiosk application with Dijkstra-based pathfinding, signage customization, and service requests over SQL backend
- Led 11-member Agile team in design, implementation, and user testing phases with modular MVC and OOP design patterns.

TECHNICAL SKILLS

Perception & Sensing	Intel Realsense, LiDAR, Image Segmentation, Object Detection, Kalman Filters, YOLOv11
ML / RL Tools	PyTorch, Scikit-Learn, Nvidia IsaacLab, DQN, Imitation Learning, ResNet, Pandas, W&B
Motion Planning	ORCA, SLAM, (gmapping), AMCL, A*, Pure Pursuit, Monocular Depth Estimation
Simulation	Nvidia IsaacSim, Gazebo, Unity Game Engine, MuJoCo, Blender (Synthetic Data Generation), PyGame
Embedded Systems	MicroPython, Arduino, Raspberry Pi, Nvidia Jetson, UART/I^{2}C, ROS/ROS2, HALs
Version Control & Dev Tools	Git, Agile, Kanban, Azure DevOps, Jira, Figma, Github Projects, Microsoft Office
Languages	Python, TypeScript, C++, C#, Java, C, JavaScript, Kotlin, MATLAB

LEADERSHIP AND EXTRACURRICULARS

Scouting America, Troop 106, Eagle Scout — **Mar 2014 – July 2021**
WPI Robotics Prototyping Club, Founder, Treasurer — **Aug 2024 - May 2025**
WPI Cooking Club, President — **Apr 2023 - May 2025**
Rho Beta Epsilon Robotics Engineering Honor Society, Alpha Chapter — **Feb 2025 - May 2025**
WPI Bowling Club, Treasurer — **Aug 2022 – Feb 2023**
WPI VexU, Software Co-Lead — **Aug 2022 – Feb 2023**